**d. Amensalism:** It is an interspecific interaction in which one species is inhibited while the other species is neither benefitted nor harmed. The inhibition is achieved by the secretion of certain chemicals called **allelopathic** substances. Amensalism is also called **antibiosis.**

- *Penicillium notatum* produces penicillin to inhibit the growth of a variety of bacteria especially *Staphylococcus*.
- *Trichoderma* inhibits the growth of fungus *Aspergillus*.
- Roots and hulls of Black Walnut *Juglans nigra* secretes an alkaloid **Juglone** which inhibits the growth of seedlings of Apple, Tomato and *Alfalfa* around it.

## Interspecific interactions/ Co-evolutionary dynamics

**i. Mimicry:** It is a phenomenon in which living organism modifies its form, appearance, structure or behavior and looks like another living organism as a self defence and increases the chance of its survival. Floral mimicry is for usually inviting pollinators but animal mimicry is often protective. Mimicry is a result of evolutionary significance due to shape and sudden heritable mutation and preservation by natural selection.

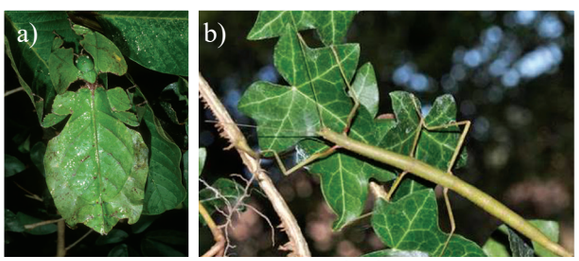



**Figure 6.15:** Mimicry
a) *Phyllium frondosum* b) *Carausium morosus*

Example:

- The plant, *Ophrys* an orchid, the flower looks like a female insect to attract the male insect to get pollinated by the male insect and it is otherwise called 'floral mimicry '.
- *Carausium morosus* – stick insect or walking stick. It is a protective mimicry.
- *Phyllium frondosum* – leaf insect, another example of protective mimicry.

**ii. Myrmecophily:** Sometimes, ants take their shelter on some trees such as Mango, Litchi, Jamun, *Acacia* etc. These ants act as body guards of the plants against any disturbing agent and the plants in turn provide food and shelter to these ants. This phenomenon is known as Myrmecophily. Example: Acacia and acacia ants.

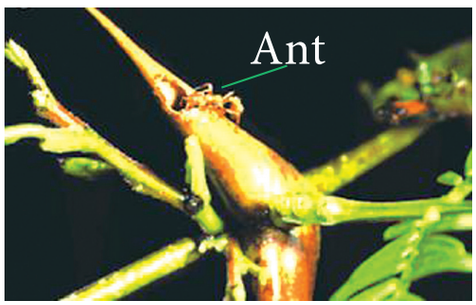



**Figure 6.16:** Myrmecophily

**iii. Co-evolution:** The interaction between organisms, when continues for generations, involves reciprocal changes in genetic and morphological characters of both organisms. This type of evolution is called Co-evolution. It is a kind of co- adaptation and mutual change among interactive species.



**Figure 6.17:** Co-evolution

Examples:

- Corolla length and proboscis length of butterflies and moths ( *Habenaria* and Moth ).
- Bird's beak shape and flower shape and size.
- More examples: Horn bills and birds of Scrub jungles ,Slit size of pollinia of Apocynaceae members and leg size of insects.



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